

Lab # 05

Aggregate Functions & DML Queries



National University
of computer and emerging sciences

Database Systems Lab (CL-2005)

Semester: Spring 2026

Section: BCS-4E

Course Instructor: Mr. Abdullah Shaikh

LAB # 05 - TASKS

Aggregate Functions & DML Queries

Submission Instructions:

- Perform all the assigned tasks carefully.
- Prepare your solutions in a Microsoft Word document, clearly writing the solution of each problem.
- Attach relevant screenshots for every task as proof of execution.
- Ensure that your document is properly formatted (clear headings, proper spacing, and alignment).
- After completing the Word document, convert it into a PDF file.
- Upload the PDF file on Google Classroom (GCR) within the given deadline.
- **No submission will be accepted after the deadline.**
- Before uploading, rename your PDF file using the following format:

DB Systems Lab 05 – Your Name – Your Roll Number

Example: DB Systems Lab 05 – Abdullah Shaikh – 25K-7826

PART - A

1. Display the total number of employees in the company.
2. Show the average salary of all employees.
3. Display the number of employees working in each department.
4. Find the maximum commission percentage given in each department.
5. Display the minimum salary offered for each job role.
6. Show the total salary paid to employees for each job_id.
7. Display the number of employees hired in each year (use hire_date).
8. Show the average commission percentage for each department, but only include departments where the average commission is greater than 0.10.
9. Display the number of employees working under each manager, but only show managers who manage more than 2 employees.
10. Show the maximum salary in each department, but only include departments where the maximum salary is greater than 12000.

PART - B

```
CREATE TABLE Students (  
    student_id NUMBER PRIMARY KEY,  
    student_name VARCHAR2(50),  
    department VARCHAR2(50),  
    semester NUMBER  
);
```

```
CREATE TABLE Courses (  
    course_id NUMBER PRIMARY KEY,  
    course_name VARCHAR2(50),  
    credit_hours NUMBER  
);
```

```
CREATE TABLE Enrollments (  
    enroll_id NUMBER PRIMARY KEY,  
    student_id NUMBER,  
    course_id NUMBER,  
    grade VARCHAR2(2),  
    CONSTRAINT fk_student  
        FOREIGN KEY (student_id) REFERENCES Students(student_id),  
    CONSTRAINT fk_course  
        FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
);
```

Note:

For the tasks given in Part B, you are required to first create the database schema using the DDL commands provided above. Please copy and paste the schema in Oracle SQL Developer and execute it to create the required tables. After successfully creating the tables, write and execute the SQL queries for each of the tasks listed in Part B. Ensure that your queries are written based on the provided schema.

Lab # 05

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Write SQL statements for the following:

1. Insert a student with ID 101, name Ali Khan, department CS, semester 3.
2. Insert another student with ID 102, name Sara Ahmed, department AI, semester 2.
3. Insert a course Database Systems with course_id 501 and credit hours 3.
4. Insert a course Operating Systems with course_id 502 and credit hours 4.
5. Insert an enrollment where student 101 enrolls in course 501 with grade 'A'.
6. Update the semester of student 102 to semester 3.
7. Update the credit hours of course 501 to 4.
8. Update the grade of student 101 in course 501 to 'A+'.
9. Delete the student whose student_id = 102.
10. Delete the course whose course_id = 502.

THE END!